

Hypothesis Testing Practical Assignment

Z-Test, t-Test & Chi-Square Test

Assignment Type: Practical Activity

Significance Level: $\alpha = 0.05$

Instructions

For **every question**, complete the following tasks:

1. Identify the most appropriate statistical test.
 2. State the Null Hypothesis (H_0).
 3. State the Alternative Hypothesis (H_1).
 4. Mention whether the test is:
 - One-tailed or Two-tailed
 - Parametric or Non-parametric
 5. Perform the statistical test.
 6. Report:
 - Test Statistic
 - Degrees of Freedom (if applicable)
 - p-value
 7. Compare the p-value with $\alpha = 0.05$.
 8. State your decision:
 - Reject H_0
 - Fail to Reject H_0
 9. Write the business conclusion in simple English.
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Section A

Z-Test (Known Population Standard Deviation)

Activity 1 – Soft Drink Filling Machine

A beverage company claims that its automated filling machine dispenses **500 ml** of soft drink into every bottle.

The quality assurance team randomly selects **50 bottles** from today's production.

The population standard deviation of the filling machine is known to be **8 ml**.

The average volume of the selected bottles is found to be **496.8 ml**.

Your Task

Determine whether there is sufficient statistical evidence to conclude that the machine is not filling bottles with an average volume of 500 ml.

Activity 2 – Internet Speed

An Internet Service Provider advertises that its average download speed is **100 Mbps**.

A random sample of **45 customers** reports an average download speed of **95.4 Mbps**.

The population standard deviation is known to be **12 Mbps**.

Your Task

Test whether the company's advertised claim is statistically valid.

Activity 3 – Waiting Time

A bank claims that customers wait an average of **6 minutes** before being served.

A random sample of **40 customers** shows an average waiting time of **6.8 minutes**.

Population standard deviation is **2 minutes**.

Your Task

Determine whether the average waiting time differs significantly from the bank's claim.

Activity 4 – Battery Weight

A battery manufacturer claims that every battery weighs **250 grams** on average.

A quality engineer measures **60 batteries**.

Sample mean = **247.5 grams**

Population standard deviation = **6 grams**

Your Task

Test the manufacturer's claim.

Activity 5 – Tablet Manufacturing

A pharmaceutical company claims that each tablet weighs **250 mg**.

A random sample of **100 tablets** has an average weight of **248.7 mg**.

Population standard deviation = **5 mg**.

Your Task

Determine whether production is meeting the required specification.

Section B

One Sample t-Test

Activity 1 – Student IQ

A school claims that the average IQ of its students is **100**.

A psychologist randomly selects **30 students**.

The sample has:

- Mean IQ = **106**
- Sample Standard Deviation = **12**

Your Task

Test whether the school's claim is statistically correct.

Activity 2 – Employee Working Hours

A company claims that employees work **8 hours per day** on average.

A sample of **28 employees** has:

- Mean = **8.6 hours**
- Standard Deviation = **1.2 hours**

Your Task

Determine whether employees actually work longer than claimed.

Activity 3 – Coffee Shop Spending

A coffee shop owner believes that customers spend **₹250** per visit.

Thirty-five customer bills are randomly selected.

Sample mean = ₹267

Sample standard deviation = ₹38

Your Task

Test whether customer spending differs from ₹250.

Activity 4 – Exam Scores

A university claims that the average examination score is **75 marks**.

Twenty students are selected.

Sample mean = **78.5**

Sample standard deviation = **10**

Your Task

Determine whether the average score differs significantly from 75.

Activity 5 – Delivery Time

A courier company claims that parcels are delivered within **3 days** on average.

Twenty-five deliveries are sampled.

Sample mean = **3.6 days**

Sample standard deviation = **0.9 days**

Your Task

Evaluate the company's claim.

Section C - Independent Two Sample t-Test

Activity 1 – Salary Comparison

A company wants to know whether there is a difference in the average salaries of Male and Female employees.

Datapoint - [Link](#)

Your Task

Determine whether gender has a significant effect on average salary.

Activity 2 – Teaching Methods

Two classes are taught using different teaching methods:

- Online
- Classroom

Datapoint - [Link](#)

Your Task

Determine whether the teaching methods produce significantly different average scores.

Activity 3 – Delivery Companies

Two courier companies deliver products across the same city.

Datapoint - [Link](#)

Your Task

Determine whether their average delivery times differ significantly.

Section D - Paired t-Test

Dataset for Paired T-test activities: [Link](#)

Activity 1 – Weight Loss Program

Twenty participants joined a weight loss program.

Their weights are measured:

- Before the program
- After 8 weeks

Your Task

Determine whether the program significantly reduces body weight.

Activity 2 – Blood Pressure

Patients' blood pressure is measured:

- Before medication
- After medication

Your Task

Determine whether the medicine is effective.

Activity 3 – Employee Training

Employee productivity scores are recorded:

- Before training
- After training

Your Task

Evaluate whether the training improved productivity.

Section E - Chi-Square Goodness of Fit Test

Activity 1 – Fair Dice

A die is rolled **120 times**.

Observed frequencies:

Face	Frequency
1	15
2	19
3	17
4	24
5	22
6	23

Your Task

Determine whether the die is fair.

Activity 2 – Coin Toss

A coin is tossed **200 times**.

Observed:

- Heads = 118
- Tails = 82

Your Task

Determine whether the coin is fair.

Activity 3 – Ice Cream Preference

A company expects equal preference among four flavours.

Observed customer preferences:

- Vanilla = 40
- Chocolate = 58
- Strawberry = 46
- Butterscotch = 56

Your Task

Determine whether customer preference is equally distributed.

Section F - Chi-Square Test of Independence

Activity 1 – Gender vs Winning

The marketing department collects the following data.

Gender	Won	Not Won
Male	120	80
Female	140	60

Your Task

Determine whether the winning ratio is independent on gender.

Activity 2 – Smoking vs Lung Disease

Data collected from a hospital.

Smoking Status	Lung Disease	No Lung Disease
Smoker	82	38
Non-Smoker	41	89

Your Task

Determine whether smoking and lung disease are associated.
